

F-VIS: Efficient Robotic Tactile Sensing via Fiber Bragg Grating-Based Force and Vision Fusion

Qiaokang Liang¹, Senior Member, IEEE, Xiaoyu Yuan, Jianyong Long², Member, IEEE, and Dan Zhang³, Senior Member, IEEE

Abstract—This article introduces F-VIS, a novel visual-tactile sensor that integrates direct force measurement based on fiber Bragg grating (FBG) with vision-based tactile sensing. F-VIS achieves multimodal tactile perception in real time by combining high-precision force and visual data, without relying on marker layers or data-driven force estimation. The sensor uses three spiral structures for 3-D force measurement (F_x , F_y , and F_z) and a designed contact body for tactile image acquisition and depth reconstruction. Static calibration experiments show that F-VIS has root-mean-square error (RMSE) values of 0.01, 0.018, and 0.022 N for F_x , F_y , and F_z within the ± 5 and $[0, 10]$ N ranges. It also achieves an average RMSE of 0.12 mm in depth reconstruction. Fusion experiments show the sensor achieves RMSE of 2.5 pixels and 0.12 N for contact location and force, respectively. With high sensitivity (up to 24.95 pm/N) and a 100-Hz operating frequency, F-VIS enables rapid, precise force detection. Temperature compensation keeps FBG drift under 7 pm ($25^\circ\text{--}50^\circ$); after 500 cycles, transmittance rises 0.36%. F-VIS excels in hardness detection, object classification, and delicate manipulation, detecting 0.16 N forces and capturing slip forces. It outperforms existing sensors with superior precision, responsiveness, and versatility.

Index Terms—Depth reconstruction, direct force measurement, fiber Bragg grating (FBG), visual-tactile sensor, without marker layers.

I. INTRODUCTION

FORCE and tactile sensing enable sensory devices to obtain information through direct contact with objects, which is crucial for robots to perform dexterous grasping [1], precise manipulation [2], and safety [3]. Vision-based tactile sensors capture tactile information by using cameras to acquire images. These sensors can not only intuitively obtain the texture of the object in contact [4] but also process the images to extract contact force information and perform depth reconstruction [5]. However, existing vision-based tactile sensors rely on the movement of a marker layer for force measurement. The introduction of the marker layer may obscure important image features [6], and due to factors, such as imaging conditions and camera frame rate, this indirect method of force estimation struggles to provide high-accuracy and real-time force estimates. Therefore, designing a vision-based tactile sensor that integrates high-quality texture information, high-quality depth information, and high-precision, high-response multidimensional force information is of significant importance for efficient human-robot interaction and dexterous robot control.

In recent years, vision-based tactile sensors have garnered widespread attention from scholars and researchers. Adelson and Johnson [4] were the first to introduce photometric stereo for surface texture reconstruction and proposed the GelSight sensor [7] through a lookup table (LUT) method that correlates color data with the object surface normal. Based on this technology, scholars and researchers have conducted in-depth studies on various applications, such as clothing attribute classification [8], fabric defect detection [9], flexible material quality evaluation [10], and material roughness [11]. In addition to photometric stereo, DenseTact [12] and DTact [13] have achieved high-resolution 3-D reconstruction through a design without a marker layer. Cui et al. [14] used a binocular camera to achieve 3-D reconstruction by analyzing the refraction of light, while the authors in [15] and [16] proposed innovative depth estimation neural networks for depth reconstruction in GelStereo [17] and GelSlim 3.0 [18] sensors. Although these studies have made significant progress in texture detection and depth reconstruction for vision-based tactile sensors, research on high-precision force measurement remains lacking. To achieve high-precision force measurement, researchers have proposed marker-layer-based force estimation methods that rely on finite element modeling and deep learning. For example, studies, such as GelSight and GelSlim, have employed black marker arrays

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Qiaokang Liang, Xiaoyu Yuan, and Jianyong Long are with the National Engineering Research Center of Robot Vision Perception and Control, School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China (e-mail: qiaokang@hnu.edu.cn; xiaoyu2023@hnu.edu.cn; ljy_cv@hnu.edu.cn).

Dan Zhang is with the Department of Mechanical Engineering, The Hong Kong Polytechnic University, Hong Kong (e-mail: dan.zhang@polyu.edu.hk).

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to estimate the dense distribution of normal and shear forces on the sensor surface [19], [20]. However, the use of permanent markers may obscure important image features [6].

Furthermore, for vision-based tactile sensors with marker layers, it is difficult to separate force and tactile information sources during calibration due to the influence of contact geometry during measurements [21]. Regardless of the force estimation method used, these methods are data-driven, but vision-based tactile sensors undergo performance changes after repeated contact, making it difficult to acquire large training datasets. Many scholars and researchers have noted that, in simulated environments, a large number of synthetic images can be generated for neural network training, which can then be transferred to real sensors. Ding et al. [22] used the Unity physics engine to simulate the deformation of TacTip sensors and then transferred the learning from the simulation to real-world scenarios, achieving submillimeter accuracy in edge prediction tasks. Subsequently, researchers developed simulators, such as Taxim [23], Tacto [24], Tactile Gym [25], and SimTacLS [26], to support a wider range of vision-based tactile sensor types. Despite these advancements, the specificity of vision-based tactile sensor simulation software and the lack of a universal simulation-to-reality methodology across different sensors have led to high research and development costs when customizing visual-tactile sensors. Therefore, developing new force measurement technologies to reduce reliance on marker layers has become a current research focus.

To achieve high-precision force sensing while minimizing the impact of the marker layer on high-quality texture acquisition, scholars and researchers have proposed a series of methods. Abad and Ranasinghe [6] addressed this issue by using UV ink and UV LEDs for marking and illumination, enabling the marker layer only when necessary to perform force sensing without compromising texture information. Li et al. [27] introduced an innovative F-TOUCH sensor, which incorporates an internal elastic structure, allowing the camera to simultaneously capture both mechanical deformation and surface deformation of the elastomer layer to estimate triaxial forces. This approach was further extended to the measurement of six-axis forces/torques [21] and shape recognition [28]. Subsequently, Li et al. [29] further optimized the optical system of the sensor, proposing a more lightweight and cost-effective version, the L3 F-TOUCH. Xu et al. [30] proposed a multimodal tactile information sensing strategy that relies solely on a single reflective optical field, eliminating the need for a marker layer. Although these methods have made progress in improving the force measurement performance of vision-based tactile sensors, they primarily rely on force estimation through cameras rather than direct force measurement. This, to some extent, limits the further enhancement of force measurement performance in vision-based tactile sensors. Multiaxis force/torque sensors in robotics are typically based on strain gauges [31], capacitive [32], magnetic [33], or optical [34] sensing. These sensors face challenges, such as optical performance compromise, in embedded configurations and poor shear sensitivity or EMI issues in detached ones. FBG-based sensors, however, offer high sensitivity, EMI immunity, compact size, and stability [35] and have been applied in various fields [36], [37], [38], [39], [40],

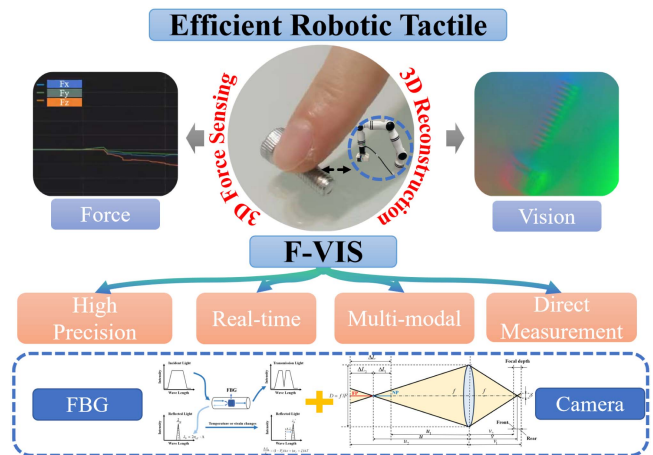


Fig. 1. Implementation framework of F-VIS.

[41], [42]. Yet, integrating FBG sensors with vision-based tactile sensors is underexplored due to their different technical principles, posing challenges, such as signal processing differences and mechanical compatibility issues.

We introduce the F-VIS sensor, which combines direct force measurement with visual information acquisition. This sensor is capable of directly measuring multimodal tactile information in real time and effectively, including force, contact area, and depth, without relying on marker layers or data-driven force estimation methods. Specifically, in this study, we have designed an ingenious elastic structure to utilize FBG for the measurement of 3-D forces (F_x , F_y , and F_z); the principle of FBGs is shown in the lower left corner of Fig. 1. Compared with existing FBG-based force-sensing studies, our elastic structure innovatively incorporates a distributed rectangular cross-section spring design. This enables the FBG fibers to sensitively respond to changes in 3-D forces while ensuring that visual imaging remains unaffected. Through model analysis, the impact of deformation on visual imaging is mitigated. The experimental results demonstrate that the F-VIS sensor has excellent performance, providing a superior solution for high-precision, real-time, multimodal, and direct measurement of tactile information.

The rest of this article is organized as follows. Section II introduces the sensor's design, operational principle, and simulation. Section III presents static calibration, dynamic experiments, and performance comparison with other tactile sensors based on visuals. Finally, Section IV concludes this article and future research prospects.

II. STRUCTURAL DESIGN AND OPERATING PRINCIPLE OF OUR SENSOR

A. Structural Design of F-VIS

This section systematically delineates the four core modules of the F-VIS (the contact module, the illumination module, the camera module, and the force-sensing module), along with supplementary components. These integrated subsystems operate synergistically to fulfill the sensor's multifunctional requirements, with the architectural configuration visualized in Fig. 2(a).

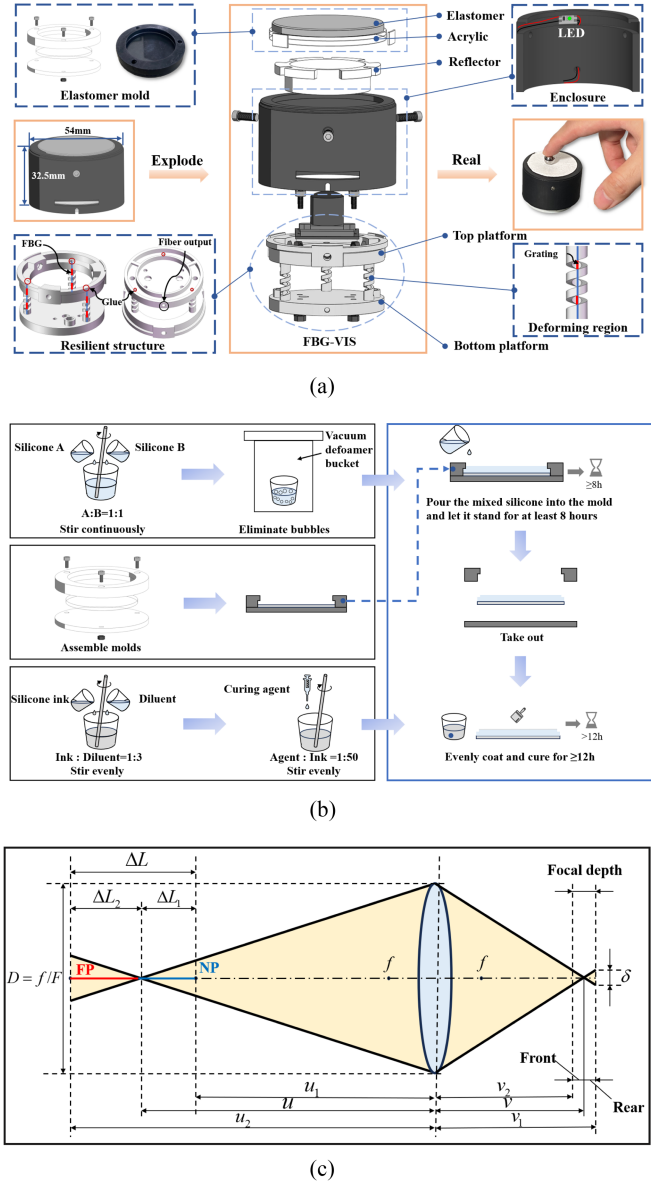


Fig. 2. (a) Sensor structure and details. (b) Manufacturing steps for the contact body and the reflective layer of the sensor. (c) Working principle of lens imaging.

- 1) **Contact module:** Comprising a transparent silicone contact body, a white silicone ink reflective layer, and a transparent acrylic plate, the contact body features Shore 10 A hardness and 400% elongation at break, with dimensions of 4.5 mm thickness and 22 mm diameter. To prioritize texture characterization, the marker layer was intentionally omitted. The reflective layer forms a uniform white film, providing essential conditions for height reconstruction, while the acrylic plate ensures structural support. The contact body was fabricated through mold-casting of degassed silicone followed by curing. The reflective layer was created by applying a mixture of diluted silicone ink and hardening agent onto the cured transparent silicone, followed by secondary curing. The manufacturing process is detailed in Fig. 2(b).

- 2) **Lighting module:** A tri-color LED array (SMD 2835, 0.1 W nominal power, 120° beam angle) with red, green, and blue channels enables side illumination. The contact body serves as an optical waveguide for 3-D reconstruction.
- 3) **Camera module:** A Waveshare IMX219 camera with a 200° ultra-wide-angle fisheye lens (0.87 mm focal length, 1/4-inch CMOS sensor) was employed. The compact module measures 25 mm×24 mm with a lens height of 13 mm, achieving 800×480 pixel resolution at 60 fps.
- 4) **Force sensing module:** Designed as an elastic structure for detecting microdeformations under load, this module incorporates grooves and threaded holes on its platform for secure mounting of reflective panels and housing integration. Three spring structures arranged at 120° intervals surround a centrally positioned FBG for deformation monitoring. The FBG installation method is illustrated in the lower left diagram of Fig. 2(a). First, the optical fiber is threaded through the fiber exit channel of the compliant structure, passed along the central axis of the hollow spring through the bottom through-hole, and then threaded out through the top through-hole, ensuring the FBG sensing area is positioned at the spring's center. Subsequently, UV adhesive is applied to secure the fiber base to the compliant structure. After pretensioning the upper fiber section, the top end is similarly fixed with UV adhesive. The base contains threaded holes for camera/sensor fixation and fiber optic guidance channels. Due to inevitable manufacturing errors introduced during the fabrication process of the elastic structure, inconsistent strain transfer may occur when applying the same force, leading to deviations in the force–wavelength mapping. To minimize such errors, this study employs a high-precision CNC machining scheme using aluminum alloy for the elastic structure.
- 5) **Other components:** The sensor housing and reflective panels were manufactured through fused deposition modeling and assembled using M2 screws. To optimize optical performance, the camera working distance was calibrated to 14 mm, resulting in a total sensor height of 32.5 mm.

B. Principle of Sensor Imaging

The F-VIS sensor uses fisheye lenses to achieve an expanded field of view and manages depth of field (DOF) to ensure sharp imaging despite variations in imaging distance due to contact surface deformation. Based on the lens imaging relationship $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, when the camera working distance is set to 14 mm, the front focal depth is $\Delta L_1 = 1.16$ mm, the rear focal depth is $\Delta L_2 = 1.39$ mm, and the total DOF reaches 2.55 mm. For ultra-wide-angle fisheye lenses, the rear focal depth is considered infinite. Camera calibration is performed using a 5×8 chessboard pattern and MATLAB's camera calibration toolbox to correct radial distortion from the fisheye lens and pincushion distortion from the acrylic plate and silicone elastomer. The photometric stereo technique maps pixelwise

luminance variations to surface normals, estimating surface gradient using RGB channels and accounting for sensor-wide color variations and nonuniform illumination artifacts. The reflection function of the F-VIS contact body is empirically calibrated, and neural networks are used instead of LUTs to generate continuous outputs, as proposed by Xu et al. [10].

C. Principle of Contact Force Measurement

The F-VIS sensor uses FBGs for force sensing, leveraging their sensitivity and EMI resistance. The sensor operates by reflecting light at the Bragg wavelength λ_b , which is modulated by the refractive index

$$\lambda_b = 2n_{\text{eff}} \cdot \Lambda \quad (1)$$

where n_{eff} is the effective refractive index of the optical fiber and Λ is the grating period. Strain or temperature changes modify the grating period and refractive index, causing a detectable change $\Delta\lambda_b$ in the Bragg wavelength. Given the focus on force-vision fusion, temperature is maintained at ambient conditions, simplifying the relationship to

$$\frac{\Delta\lambda_b}{\lambda_b} = (1 - \rho_e) \cdot \Delta\varepsilon \quad (2)$$

where $\Delta\varepsilon$ is the axial strain and ρ_e is the photoelastic coefficient. When the optical fiber attached to the compliant structure is subjected to external force (tension or compression), the grating period Λ changes, leading to a shift in the Bragg wavelength λ_b . By measuring the variation in Bragg wavelength $\Delta\lambda_b$, the external force or strain can be determined.

When an elastic material undergoes a normal force F_z , its deformation can be modeled as a spring. For a helical spring with a rectangular cross section, h and w are the height and width of the cross section, respectively, and L is the diameter of the helical structure, as shown in Fig. 3(a). The stiffness K_s of a rectangular cylindrical cross section spring is calculated as follows:

$$K_s = \frac{64Gh^3w}{N\pi L^3} \left[\frac{1}{3} - \frac{64h}{\pi^5 w} \sum_{n=1}^{\infty} \frac{1}{n^5} \tanh\left(\frac{nw\pi}{2h}\right) \right] \quad (3)$$

where G is the shear modulus and N represents the number of active coils in the spring, and generally, take $n = 1, 3, 5$. For three rectangular cross-section hollow cylindrical springs, if pressed vertically at the center, F_z is applied at the geometric center of the cylinder. According to classical mechanics principles, a force applied through the centroid will produce a pure axial stress state; the deformation Δz of the elastic material can be expressed as follows:

$$\Delta z = \frac{F_z}{3K_s + 3K_f} \quad (4)$$

where K_f represents the stiffness of the Fiber Bragg Grating. The derived (4) aligns well with the simulation results shown in Fig. 3(a). Specifically, when the compliant structure is subjected to a force along the negative z -axis direction, all three FBGs experience simultaneous compressive strain.

When the elastic material is subjected to radial forces, the bending deformation induces strain in the FBG. As shown in

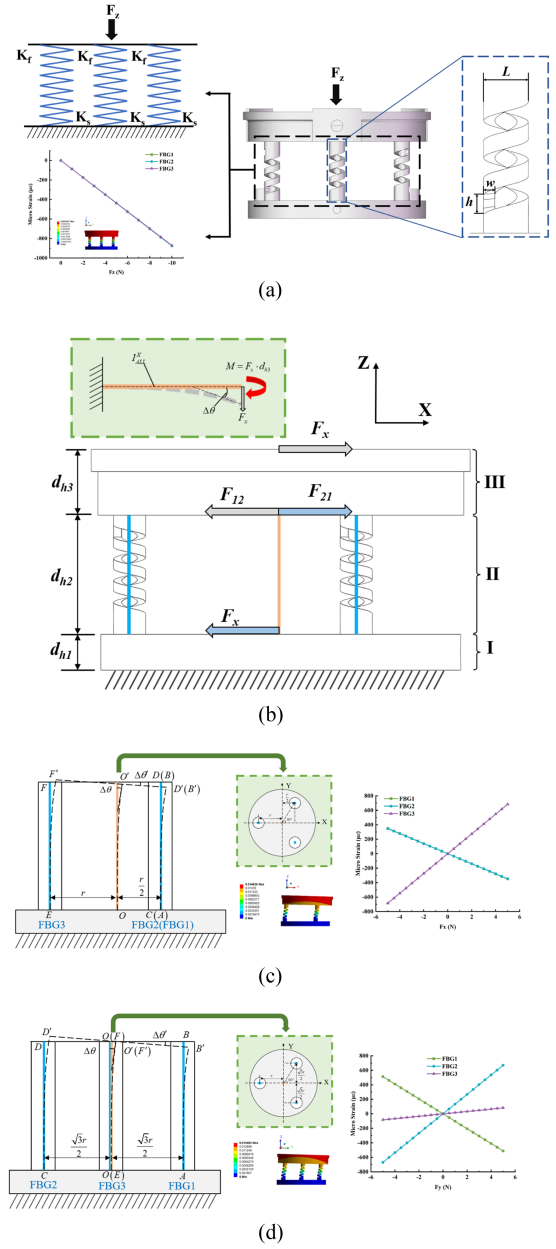


Fig. 3. (a) Deformation principle and finite element simulation of the elastic structures under normal force. (b) Deformation principle of the elastic structures under radial force. (c) and (d) Mechanical analysis and finite element simulation of an elastic structure under radial force, respectively.

Fig. 3(b), the structure is divided into three segments, with segments I and III approximated as rigid and segment II modeled as a cantilever beam. The deformation of the FBGs is approximated as an arc structure. When subjected to x -axis forces, by mapping the deformation of the three FBGs onto the neutral surface, the geometric relationships yield

$$\Delta\theta' = \Delta\theta = -\frac{F_x d_{h2}(d_{h2} + 2d_{h3})}{2EI_y^{\text{ALL}}} \quad (5)$$

where F_x is the force applied in the x -direction, I_y^{ALL} represents the equivalent moment of inertia of the entire elastic structure on the y -axis, E represents the Young modulus of the elastic

structure, d_{h2} represents the length of the spring structure, and d_{h3} represents the distance from the point of application of force to the connection area between parts II and III.

As shown in Fig. 3(c), assuming that the radius of curvature of $\widehat{OO'}$ is R and the distance from the spring structure to the center of the entire elastic structure is r . Then, the radii of $\widehat{AB'}$, $\widehat{CD'}$, and $\widehat{EF'}$ are $R - r/2$, $R - r/2$, and $R + r$, respectively. The average strain $\Delta\varepsilon_{\text{FBG1}}$ of $\widehat{AB'}$ can be expressed as follows:

$$\Delta\varepsilon_{\text{FBG1}} = \frac{\widehat{AB'} - \widehat{OO'}}{\sum_{i=1}^3 d_{hi}} = -\frac{rF_x d_{h2}(d_{h2} + 2d_{h3})}{4EI_y^{\text{ALL}} \sum_{i=1}^3 d_{hi}} \quad (6)$$

where d_{h1} represents the height of the bottom structure.

Similarly, the average strains of $\widehat{CD'}$ and $\widehat{EF'}$ can be expressed as follows:

$$\Delta\varepsilon_{\text{FBG2}} = \frac{\widehat{CD'} - \widehat{OO'}}{\sum_{i=1}^3 d_{hi}} = -\frac{rF_x d_{h2}(d_{h2} + 2d_{h3})}{4EI_y^{\text{ALL}} \sum_{i=1}^3 d_{hi}} \quad (7)$$

$$\Delta\varepsilon_{\text{FBG3}} = \frac{\widehat{EF'} - \widehat{OO'}}{\sum_{i=1}^3 d_{hi}} = \frac{rF_x d_{h2}(d_{h2} + 2d_{h3})}{2EI_y^{\text{ALL}} \sum_{i=1}^3 d_{hi}}. \quad (8)$$

When the elastic structure is subjected to a load in the positive y -direction, as shown in Fig. 3(d), the average strains of the respective FBGs can be expressed as follows:

$$\Delta\varepsilon_{\text{FBG1}} = \frac{\widehat{AB'} - \widehat{OO'}}{\sum_{i=1}^3 d_{hi}} = -\frac{\sqrt{3}rF_y d_{h2}(d_{h2} + 2d_{h3})}{4EI_x^{\text{ALL}} \sum_{i=1}^3 d_{hi}} \quad (9)$$

$$\Delta\varepsilon_{\text{FBG2}} = \frac{\widehat{CD'} - \widehat{OO'}}{\sum_{i=1}^3 d_{hi}} = \frac{\sqrt{3}rF_y d_{h2}(d_{h2} + 2d_{h3})}{4EI_x^{\text{ALL}} \sum_{i=1}^3 d_{hi}} \quad (10)$$

$$\Delta\varepsilon_{\text{FBG3}} = \frac{\widehat{EF'} - \widehat{OO'}}{\sum_{i=1}^3 d_{hi}} = 0. \quad (11)$$

Equations (6)–(11) also align well with the finite element analysis results shown in Fig. 3(c) and (d). Specifically, when the compliant structure is loaded along the positive x -axis, FBG1 and FBG2 experience compression with identical strain magnitudes, while FBG3 undergoes tension with approximately twice the strain magnitude of FBG1/FBG2 in the opposite direction. Under loading along the positive y -axis, FBG1 experiences compression and FBG2 tension with equal-but-opposite strain magnitudes, whereas FBG3 shows negligible strain.

Assuming that the central wavelength of the optical fiber is λ_B , by substituting the normal force-induced strain (4) and the tangential force-induced strains (6)–(11) into the FBG wavelength shift–strain relationship formula (2), the FBG central wavelength shift can be derived as follows:

$$\begin{cases} \Delta\lambda_1 = C_1 \cdot (-A_1 \cdot F_x - \sqrt{3}A_2 F_y + B \cdot F_z) \\ \Delta\lambda_2 = C_1 \cdot (-A_1 \cdot F_x + \sqrt{3}A_2 F_y + B \cdot F_z) \\ \Delta\lambda_3 = C_1 \cdot (2A_1 \cdot F_x + B \cdot F_z) \end{cases} \quad (12)$$

where $A_1 = \frac{rd_{h2}(d_{h2}+2d_{h3})}{4EI_y^{\text{ALL}} \sum_{i=1}^3 d_{hi}}$, $A_2 = \frac{rd_{h2}(d_{h2}+2d_{h3})}{4EI_x^{\text{ALL}} \sum_{i=1}^3 d_{hi}}$, $B = \frac{1}{3K_s+3K_f}$, and $C_1 = \lambda_B \cdot (1 - \rho_e)$. Equation (12) expressed in

matrix form as follows:

$$\begin{bmatrix} \Delta\lambda_1 \\ \Delta\lambda_2 \\ \Delta\lambda_3 \end{bmatrix} = C_1 \cdot \begin{bmatrix} -A_1 & -\sqrt{3}A_2 & B \\ -A_1 & \sqrt{3}A_2 & B \\ 2A_1 & 0 & B \end{bmatrix} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}. \quad (13)$$

To address the coupling characteristics of fiber wavelength shift and force loading, as well as challenges from theoretical model inaccuracies and calibration errors, we adopted an extreme learning machine (ELM) network [41]. Its nonlinear mapping capability enables effective extraction of valuable information from coupled data, achieving rapid training convergence without complex gradient descent and backpropagation processes.

In the ELM framework, we establish a mapping between input and output data where the following holds.

- 1) *Input vector* $\mathbf{x}_i = [\Delta\lambda_{\text{FBG1}_i}, \Delta\lambda_{\text{FBG2}_i}, \Delta\lambda_{\text{FBG3}_i}]^T$ represents the wavelength shifts of three FBGs during calibration.
- 2) *Output vector* $\mathbf{t}_i = [\Delta F_{x_i}, \Delta F_{y_i}, \Delta F_{z_i}]^T$ denotes the tri-axial force components applied during sensor calibration loading.

Assuming that we have collected m calibration datasets with n denoting the number of hidden layer neurons, the hidden layer output of the ELM for each input sample can be expressed as follows:

$$\mathbf{h}_i = g(\mathbf{W}\mathbf{x}_i + \mathbf{b}) \quad (14)$$

where $\mathbf{W} \in \mathbb{R}^{n \times 3}$ denotes the input-to-hidden weight matrix, and $\mathbf{b} \in \mathbb{R}^{n \times 1}$ represents the hidden layer bias vector. These parameters remain fixed during training. The nonlinear activation function $g(\cdot)$ (e.g., Sigmoid, tanh, or ReLU) generates the hidden layer output $\mathbf{h}_i \in \mathbb{R}^{n \times 1}$ for each sample. The aggregated hidden layer output matrix $\mathbf{H} \in \mathbb{R}^{n \times m}$ is constructed by stacking outputs from m samples.

The output layer weight matrix $\beta \in \mathbb{R}^{n \times 3}$ can be computed via the following formula:

$$\beta = \mathbf{H}^\dagger \mathbf{T} \quad (15)$$

where $\mathbf{H}^\dagger \in \mathbb{R}^{n \times m}$ is the Moore–Penrose pseudoinverse of $\mathbf{H}$, and $\mathbf{T} \in \mathbb{R}^{m \times 3}$ denotes the target output matrix. This analytical solution method computes β directly via least squares without requiring iterative optimization.

III. CALIBRATION AND EXPERIMENTATION

A. Calibration of F-VIS Force Performance

A dedicated force calibration system, illustrated in Fig. 4, was developed to establish a reliable relationship between applied loads and the center wavelength shift of the FBG for our F-VIS sensor. For radial force calibration, the sensor was fixed on a microadjustment stage with a loading cap, and standard weights were applied via a thin wire and pulley system. For normal force calibration, weights were stacked directly on the sensor after removing the loading cap. In the static calibration experiment, the radial force range was $[-5, 5]$ N with a step size of 0.5 N, and the axial force range was $[0, 10]$ N with a step size of 1 N. Each loading and unloading cycle was repeated six times, and the average wavelength shift was recorded. Results demonstrated a

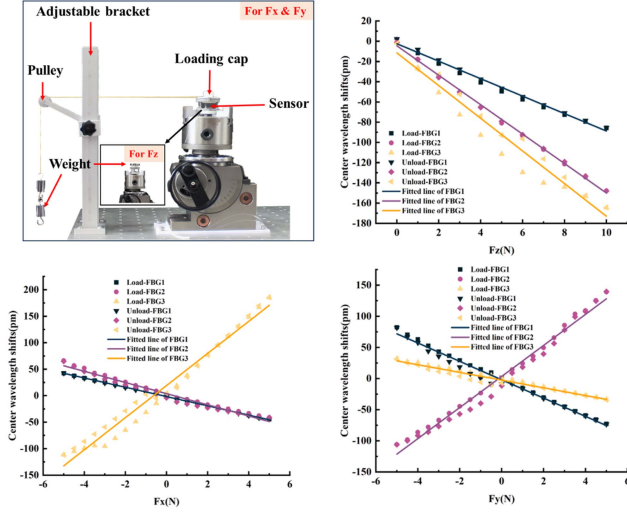


Fig. 4. Composition of the sensor calibration system and the calibration results of the x-axis, y-axis, and z-axis.

TABLE I
PERFORMANCE PARAMETER TABLE OF THE SENSOR

Force	FBG	Sensitivity (pm/N)	R-square (%)	Linearity (%)	Hysteresis (%)
F_x	FBG1	8.76	99.97	1.15	5
	FBG2	10.54	98.35	8.44	4.6
	FBG3	30.3	98.68	6.9	3.3
F_y	FBG1	14.73	99.37	6.55	5.9
	FBG2	24.95	98.89	6.25	6.5
	FBG3	6.18	99.22	6.9	3.3
F_z	FBG1	8.64	99.47	4.37	4.4
	FBG2	14.55	99.76	3.25	2.2
	FBG3	16.15	98.62	6.62	5.2

strong linear relationship between the central wavelength shift of the three FBGs and the applied load, with an R^2 value close to 1. The maximum sensitivities of FBG1, FBG2, and FBG3 were measured to be 14.73, 24.95, and 30.30 pm/N, respectively. The calibration results of the sensor [see Fig. 4(b)–(d)] demonstrate that its stress–strain relationship closely matches the simulation results presented in Section II-C. Detailed performance parameters are listed in Table I.

Building upon the ELM framework described in Section II-C, we implemented the Sigmoid activation function. The calibrated dataset was preprocessed and randomly partitioned into training and test sets with a 7:3 split ratio.

An insufficient number of hidden neurons fails to establish complex mapping relationships, while excessive neurons increase training time and may cause overfitting. To determine the optimal number of hidden layer neurons, this article investigates the relationship between decoupling time, root-mean-square error (RMSE), and hidden neuron count. The RMSE calculation formula is given by the following:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{j=1}^N (F_j - \hat{F}_j)^2} \quad (16)$$

where N denotes the number of data samples in the test set, F_j denotes the applied triaxial force components during loading, and $\hat{F}_j$ represents the network-predicted triaxial forces. We analyzed the ELM decoupling performance by adjusting the hidden

TABLE II
 R^2 , FORCE DECOUPLING RMSE, AND TRAINING TIME CORRESPONDING TO DIFFERENT DECOUPLING METHODS

Method	R^2	Average RMSE	Cost: Time (s)
Linear	0.9886	0.197	0.002
MLP	0.9993	0.051	3.023
SVR	0.9999	0.020	31.604
ELM	0.9999	0.019	0.064

TABLE III
DECOUPLING RESULTS WITH ER AND RMSE

Decoupling results	ER (%)			RMSE (N)	NRMSE (%)
	F_x	F_y	F_z		
F_x	0.311	0.242	0.048	0.017	0.340
F_y	0.119	0.421	0.049	0.018	0.357
F_z	0.133	0.260	0.201	0.022	0.219

The bold values denote the Type-I errors of the sensor.

layer neurons from 5 to 100, finding that performance plateaued beyond 50 neurons, so we selected 50 neurons for the hidden layer to balance efficiency and performance. In addition, we compared ELM with three decoupling methods: a linear model, support vector regression (SVR), and a multilayer perceptron neural network. ELM achieved the lowest average RMSE of 0.016, 0.022, and 0.018 N for the three axes, respectively, and had significantly shorter computational time than multilayer perceptron (MLP) and SVR, as detailed in Table II.

Furthermore, to comprehensively evaluate the sensor's decoupling performance metrics, we define two additional quantitative indicators: the force prediction error rate (ER) and normalized root-mean-square error (NRMSE). The ER and NRMSE can be, respectively, expressed as follows:

$$\text{ER} = \left| \frac{F_G - F_P}{F_A} \right| \times 100\% \quad (17)$$

$$\text{NRMSE} = \frac{\text{RMSE}}{F_A} \times 100\% \quad (18)$$

where F_G denotes the actual applied force, F_P represents the ELM-predicted force, and F_A indicates the full-scale value of the force sensor. The final results, listed in Table III, show the (type-I) errors for F_x , F_y , and F_z are 0.311%, 0.421%, and 0.503%, respectively, with the maximum interaxis coupling error (type-II) for F_z and F_y at 0.201%. These results indicate minimal force coupling between each dimension of the sensor after decoupling.

B. Random Force Direction Dynamic Response Experiment

To verify the sensor's multi-axis dynamic force detection capabilities, we designed a dynamic response experiment as shown in the figure. In this experiment, the sensor was subjected to a constant load along a random direction, followed by the sudden burning of a thin wire at approximately 91 s. The experimental results show that the force outputs in the three directions return to the ± 0.1 N range within about 31 ms after the cut. The response curves are illustrated in Fig. 5.

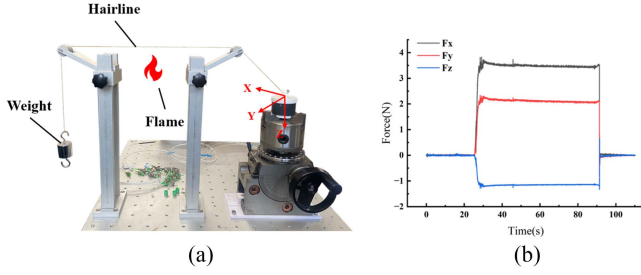


Fig. 5. (a) Experimental setup. (b) Experimental output curve.

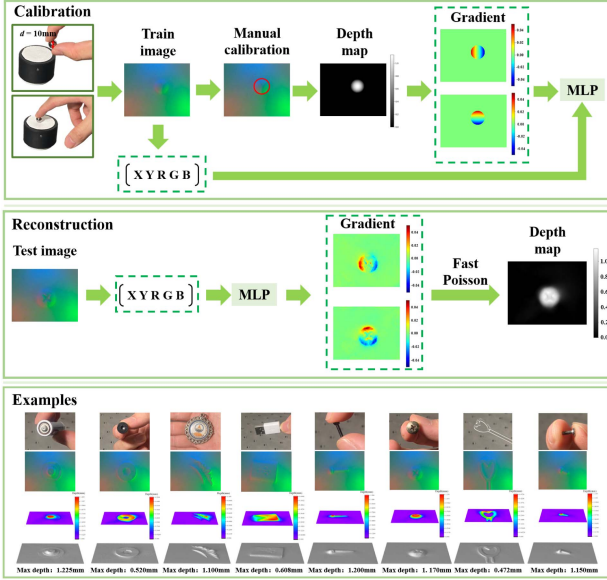


Fig. 6. Calibration and reconstruction of sensor depth reconstruction.

C. Depth Reconstruction

For tactile perception in F-VIS, depth reconstruction involves calculating surface gradients from RGB images and integrating them using a fast Poisson solver [4]. Calibration is crucial for accurate depth conversion. We used objects with known dimensions to derive pixel-to-physical size conversion and a 10 mm calibration sphere to obtain depth values and gradients G_x and G_y . To compensate for nonuniform illumination, pixel coordinates (X, Y) were incorporated alongside RGB values. For color-to-gradient mapping, an MLP with four fully connected layers and dimensions (5,32), (32,32), (32,32), and (32,2) was used, replacing traditional LUTs [8], [43]. The dataset from 120 tactile images was split into training and test sets in a 7:3 ratio. Results showed an average RMSE of 0.12 mm for the calibration sphere, with qualitative 3-D reconstructions of other objects demonstrated in Fig. 6.

D. Hardness Testing Experiment

We tested the F-VIS sensor's ability to simultaneously understand object hardness using tactile and force information on four silicone samples with hardness levels 0, 12.4, 20.7, and 33.3 A. A UR3 robotic arm pressed the samples vertically at 0.25 mm/s, increasing indentation depth in 0.5 mm increments

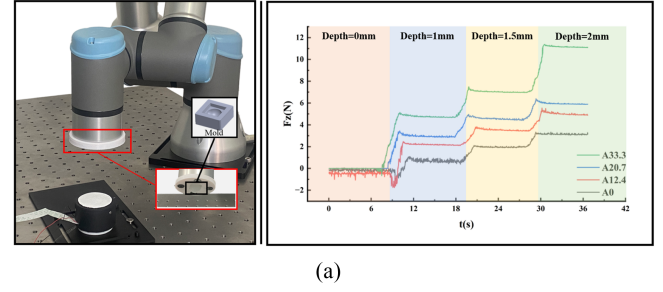


Fig. 7. (a) Left: Hardness testing experimental system composition; Right: Force information from the sensor for silicones of varying hardness at different compression depths. (b) Depth information from the sensor for silicones of varying hardness at different compression depths.

with 6-s pauses, while recording FBG strain data and tactile images. Results showed that F-VIS effectively integrates force and visual information: at 0 A hardness, tactile images were indistinct, but force data clearly indicated depth changes; at higher hardness levels (20.7 and 33.3 A), force data distinguished between levels despite similar tactile images. The sensor's output force correlated positively with material hardness at identical indentation depths, and distinct differences in indentation depths were observable in tactile images for varying hardness levels, as illustrated in Fig. 7(a) and (b).

E. Clamping Task

To demonstrate the F-VIS system's sensitivity and slip detection performance, two experiments were conducted. In the eggshell grasping experiment, eggshell fragments were positioned between gripper fingers using precision tweezers. The gripper closed until contact was detected, triggering motion cessation. The sensor captured the eggshell's morphology through tactile mapping and depth reconstruction, detecting force variations as low as 0.16 N (see Fig. 8). For slip force detection, a cylindrical test specimen ($\phi 10 \times 15$ mm) was placed between the gripper fingers of a stiffer CNC-machined aluminum gripper. Controlled perturbations along the Y -axes generated force profiles in regions I and II. The tactile imaging sequence documented contact deformation during slip events, confirming the system's ability to detect both the magnitude and directionality of slip forces (see Fig. 9).

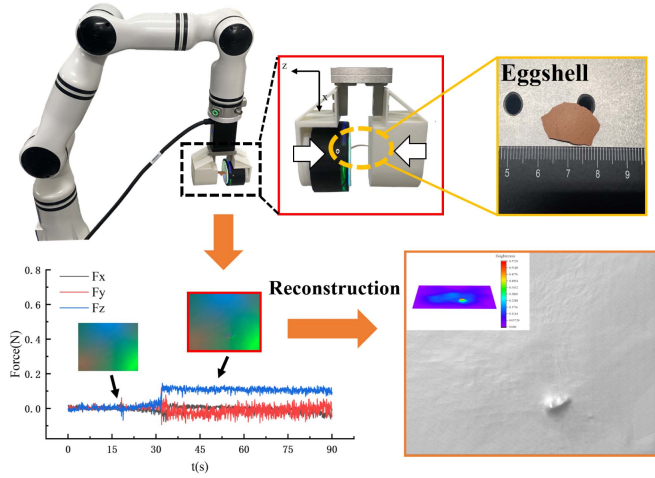


Fig. 8. Experimental outcomes of eggshell fragment grasping. The middle panel presents the force variation profiles during grasping operations, while the right panel displays postgrasping depth reconstruction results.

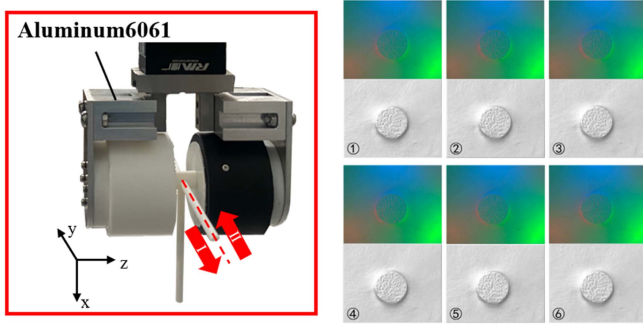


Fig. 9. System response to directional force perturbations on cylindrical specimen, showing force evolution and tactile imaging variations.

F. Multimodal Fusion for Distributed Force Perception

This study proposes fusing FBG sensor data with visual depth information for markerless distributed contact force perception. To handle the different sampling rates (FBG: $\sim 100\text{Hz}$; images: 30 fps), synchronization is achieved through hardware coordination, FBG signal filtering/downsampling, phase alignment, and label alignment. The network architecture, as shown in Fig. 10, processes images via a modified UNet [50] and FBG signals through fully connected layers, then concatenates the features. The fusion model outputs the contact region's center coordinates, radius, and total force. The dataset was split into training, validation, and test sets in a 7:1.5:1.5 ratio and trained

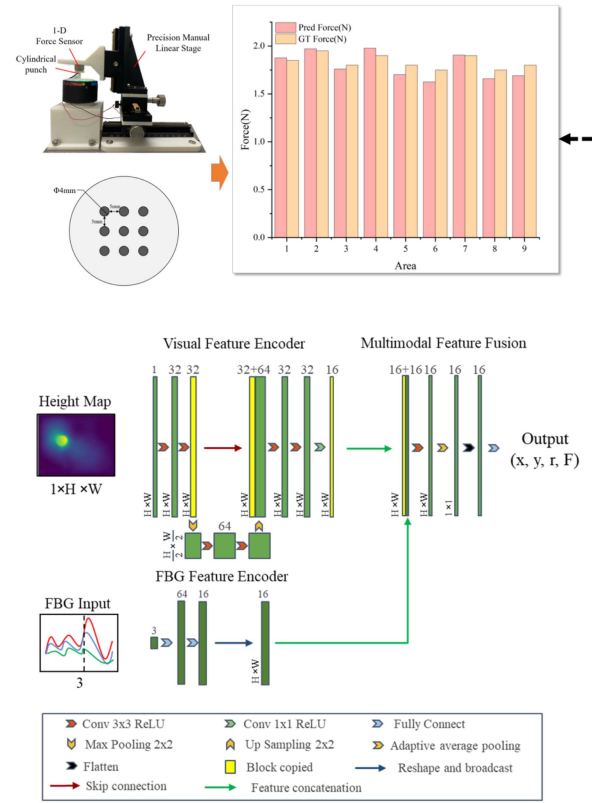


Fig. 10. Upper: Comparison of force experimental setup and results. Lower: Multimodal fusion network architecture.

using a weighted loss function as follows:

$$\mathcal{L} = \alpha \cdot \text{MSE}(\hat{x}, x) + \beta \cdot \text{MSE}(\hat{F}, F) \quad (19)$$

where $\alpha = 1.0$ and $\beta = 10.0$. The model was trained for 200 epochs using the Adam optimizer with an initial learning rate of 0.001. The results demonstrated good regional understanding and stable force prediction performance. For the test set, the RMSE of position prediction for each region is within 2.5 pixels, and the error variation between different force levels is not significant, with RMSE within 0.12 N. In addition, we tested loading at the same depth across different regions, and the experiment confirmed the model's ability to consistently measure forces at different positions with the same loading depth. The maximum error occurring in region 6 was 0.12 N. Ablation experiments compared the fusion model with single-modality models. The fusion model showed lower prediction errors, underscoring the value of multimodal information. Specifically, the total RMSE values were 3.53 for the ImageOnly model, 1.25 for the FBG-Only model, and 0.89 for the Fusion model, highlighting the superiority of the multimodal approach.

G. Comparison of Force Measurement Performance

Our F-VIS sensor demonstrates superior force measurement performance, with RMSE for both radial and normal forces being less than 0.022 N, outperforming other sensors, such as [49] (RMSE of 0.18 N for radial force and 0.15 N for normal force). Our sensor covers a broad measurement range from -5 to 5 for radial and normal forces and 0 to 10 for axial force

TABLE IV
COMPARISON OF FORCE MEASUREMENT WITH OTHER SENSORS

Reference	Input	Methods	Force	Measuring range(N)	Precision	Frequency	Measurement method
[21]	Marker displacement + Area changes	Physical model	F_x F_y F_z	-1-1 -1-1 0-8	RMSE: 0.04 N RMSE: 0.05 N RMSE: 0.13 N	30 Hz	Indirect
[44]	Optical flow	Physical model	F_x F_y F_z	-2.49-2.94 -2.82-2.86 0-9.67	RMSE: 0.14 RMSE: 0.17 RMSE: 0.30	60 Hz	Indirect
[45]	DVS image	Network model	F_n	0-20	MSE: 0.16 N	Latency:12 ms	Indirect
[29]	ARTag displacement	Physical model	F_x F_y F_z	-2-2 -2-2 0-12	RMSE: 0.2346 N RMSE: 0.1573 N RMSE: 1.33 N	26 Hz	Indirect
[46]	Marker position	Physical model	F_n	0-1	RMSE: 0.02 N	60 Hz	Indirect
[47]	Marker displacement	Physical model	F_n	0-1	RMSE: 0.03 N (DIS)	60 Hz	Indirect
[48]	UV-marked areas images + reflective film images	Physical model	F_n	0-10	RMSE: 0.264 N	90 Hz	Indirect
[49]	Image of shadow area	Physical model	F_n F_s	1.1-15.5 0-4.3	ER: 5.47% ER: 4.78%	29 Hz	Indirect
Ours (F-VIS)	FBG center wavelength shifts	Network model	F_x F_y F_z	-5-5 -5-5 0-10	RMSE: 0.017 N RMSE: 0.018 N RMSE: 0.022 N	100 Hz	Direct

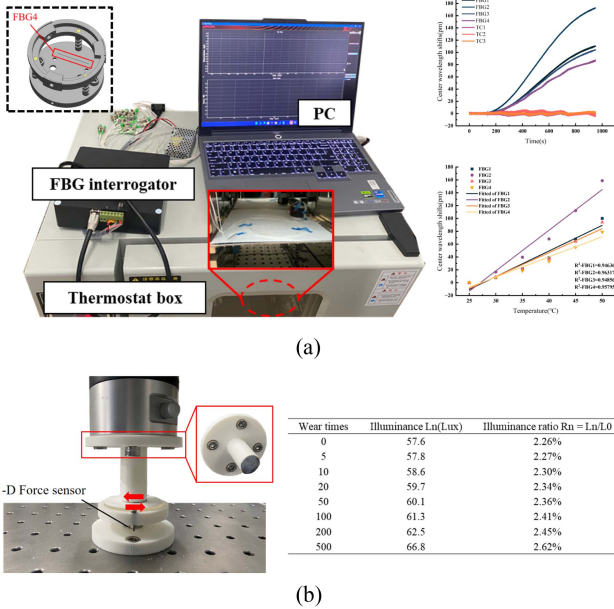


Fig. 11. (a) Temperature compensation experiment setup and results. (b) Durability test setup and results.

while operating at a frequency of 100 Hz, which is higher than most compared sensors. A detailed comparison is provided in Table IV.

Furthermore, our sensor employs a direct measurement approach based on FBG center wavelength shifts, offering several advantages over indirect methods that rely on markers and finite element or network models [21], [29], [44]. It ensures independence between force sensing and visual sensing, making the fusion of the two modalities more reliable. It also guarantees accurate force measurement under harsh environmental conditions, such as changes in ambient light, lens contamination, creep of the elastomer, or detachment or blurring of markers. In contrast, indirect methods are limited by factors, such as camera resolution, image processing precision, and material model accuracy. Errors in these areas can introduce inaccuracies, and their dynamic response is constrained by the camera's frame

TABLE V
COST COMPARISON WITH OTHER VISUO-TACTILE SENSORS

Sensor	Cost (\$)
GelSight [19]	30
GelSight-Mini	504†
DIGIT [51]	15*/350†
GelSlim3.0 [18]	25*
F-TOUCH [27]	40
L3F-TOUCH [29]	31
DAct [13]	34
F-VIS (Ours)	164

† indicates the commercial price, * indicates a bulk of 1000 units.

rate. Our sensor's direct measurement approach, combined with a well-rigid spring structure, enables high-precision, low-latency force measurement.

H. Temperature Compensation and Durability Test

To address temperature effects, a temperature-compensating FBG (FBG4) was added to the sensor. The specific implementation steps are shown in [41]. Experimental results in a temperature chamber [see Fig. 11(a)] showed that after compensation, the maximum wavelength shifts for FBG1-FBG3 were reduced to less than 7 pm. The FBG sensor's enclosed optical fiber structure minimizes humidity effects, and its inherent immunity to electromagnetic interference ensures reliable force sensing even when camera systems fail.

An automated wear test using a robotic arm with a sandpaper-covered cylindrical head was conducted. The initial transmittance of the elastomer sample was 57.6 Lux. After 500 wear cycles, the transmittance only increased by 0.36% [see Fig. 11(b)], and no obvious cracks or delamination were observed on its inner surface.

IV. CONCLUSION

This study introduces a visual-tactile sensor that uses FBG for 3-D force measurement, combining force perception with visual-tactile sensing. The sensor features three spiral structures for force perception and a designed contact body for tactile

imaging and depth reconstruction. Static calibration experiments show RMSE values of 0.017, 0.018, and 0.022 N for F_x , F_y , and F_z . Data fusion experiments achieve an RMSE of 2.5 pixels for contact location and 0.12 N for force prediction. Temperature compensation reduces FBG center wavelength drift error to below 7 pm, and the contact elastomer shows no surface damage after 500 friction cycles. The sensor's performance is verified in various tasks, including hardness detection, object classification, grasping fragile objects, and slip detection.

In future work, we aim to optimize the sensor's structure for miniaturization to enable robotic finger mounting, which is challenging in elastic structure design. We also plan to enhance the sensor for six-degree-of-freedom force measurement by optimizing the elastic structure and adding FBGs to measure contact torque. Currently, the sensor's cost is higher due to costly FBGs and demodulators, as shown in Table V. To address this, we will explore more cost-effective direct force measurement solutions to balance performance and cost, making the technology more accessible. Our ultimate goal is to develop a compact, high-precision six-degree-of-freedom visual-tactile force measurement sensor.

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Qiaokang Liang (Senior Member, IEEE) received the Ph.D. degree in control science and engineering from the University of Science and Technology of China, Hefei, China, in 2011.

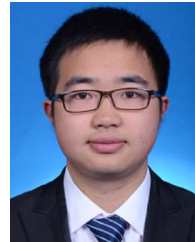
He is currently a Professor with the School of Artificial Intelligence and Robotics, Hunan University, Changsha, China, where he is also the Vice Director of the National Engineering Research Center of Robot Vision Perception and Control. His research interests include robotic perceptions, advanced robot technology, and

application of artificial intelligence technology in perceptions.



Xiaoyu Yuan received the B.S. degree in engineering from Southwest Jiaotong University, Chengdu, China, in 2023. He is currently working toward the master's degree in control science and engineering with the School of Artificial Intelligence and Robotics, Hunan University, Changsha, China.

His research interests include machine vision, tactile sensing, and robotic perception.



Jianyong Long (Member, IEEE) received the M.S. degree in agricultural information technology from the College of Information and Computer Engineering, Northeast Forestry University, Harbin, China, in 2017, and the Ph.D. degree in control science and engineering from Hunan University, Changsha, China, in 2022.

He is currently a Postdoctoral Fellow with the School of Artificial Intelligence and Robotics, Hunan University. His research interests include machine learning, robotics, and mechatronics.



Dan Zhang (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from Laval University, Quebec City, QC, Canada, in June 2000.

He is a Chair Professor of intelligent robotics and automation with The Hong Kong Polytechnic University, Hong Kong. His research interests include robotics and mechatronics, high-performance parallel robotic machine development, micromanipulation/nanomanipulation and MEMS devices, and rehabilitation robots and

rescue robots.

Dr. Zhang is a Fellow of the Canadian Academy of Engineering, the Engineering Institute of Canada, the American Society of Mechanical Engineer, and the Canadian Society for Mechanical Engineering, and a Senior Member of SME.